

The diagram illustrates the sulfur cycle with the following components and reactions:

- Species:** HS^- , $\text{S}(0)$, SO_3^{2-} , $\text{S}_2\text{O}_3^{2-}$, $\text{S}_4\text{O}_6^{2-}$, SO_4^{2-} , APS .
- Enzymes and Pathways:**
 - $\text{SO}_3^{2-} \rightarrow \text{HS}^-$: DsrMKJOP (releasing e^-).
 - $\text{HS}^- \rightarrow \text{S}(0)$: FccAB/Sqr.
 - $\text{S}(0) \rightarrow \text{SO}_3^{2-}$: SOR (releasing e^-).
 - $\text{S}(0) \rightarrow \text{S}_2\text{O}_3^{2-}$: TetH, SoxB.
 - $\text{S}_2\text{O}_3^{2-} \rightarrow \text{SO}_3^{2-}$: PhsABC.
 - $\text{S}_2\text{O}_3^{2-} \rightarrow \text{S}_4\text{O}_6^{2-}$: TsdA/DoxDA (solid), Ttr/Otr (dashed).
 - $\text{S}_4\text{O}_6^{2-} \rightarrow \text{SO}_3^{2-}$: TetH, SoxB.
 - $\text{S}_4\text{O}_6^{2-} \rightarrow \text{SO}_4^{2-}$: TetH, SoxB.
 - $\text{SO}_3^{2-} \rightarrow \text{SO}_4^{2-}$: SoeABC, SorAB, SorT.
 - $\text{SO}_4^{2-} \rightarrow \text{APS}$: Sat.
 - $\text{APS} \rightarrow \text{SO}_3^{2-}$: AprAB (releasing e^-), QmoABC.
 - $\text{S}(0) \rightarrow \text{S}_2\text{O}_3^{2-}$ (dashed): SoxYZXAB.
 - $\text{S}_2\text{O}_3^{2-} \rightarrow \text{SO}_4^{2-}$ (dashed): SoxYZXABCD.
 - $\text{S}(0) \rightarrow \text{HS}^-$ (dashed): PsrABC/ShyABCD, SudAB/Npsr/Nsr.
- Other Labels:** DsrABC, AsrABC, MccA; sHdrAB1B2B3C1C2, DsrABCEFH, Sdo.

[illegible]

Dissimilatory sulfide oxidation pathway

The diagram illustrates the dissimilatory sulfide oxidation pathway across three compartments: Periplasm, Membrane, and Cytoplasm.

- Periplasm:**
 - FccAB** (orange box) is involved in a cycle with HS^- and HS_n^- .
 - Sqr** (green box) is located in the Membrane, converting $\text{MQ} \rightarrow \text{MQH}_2$.
 - R-S-SH** (red SH) is transferred from the periplasm to the cytoplasm.
- Membrane:**
 - DsrMKJOP** (green box) is located in the Membrane, converting $\text{MQ} \rightarrow \text{MQH}_2$.
 - QmoABC** (green box) is located in the Membrane, converting $\text{MQ} \rightarrow \text{MQH}_2$.
- Cytoplasm:**
 - DsrEFH** (orange box) is involved in a cycle with HS^- and HS-SH .
 - DsrC** (blue circle) forms a disulfide bond (S-S(0)).
 - DsrBA** (yellow box) and **DsrL** (yellow box) are involved in the conversion of DsrC-SH to SO_3^{2-} .
 - AprBA** (green box) is involved in the conversion of SO_3^{2-} to **APS** (red text).
 - Sat** (yellow box) is involved in the conversion of **APS** to SO_4^{2-} (red text).

Additional details: A cycle involving S^0 (red circle) and R-S-SH is shown in the periplasm. A dashed arrow indicates electron flow (e^-) from **QmoABC** to **AprBA**.